GOOGLE DEEPMIND · OCT 2026

Meet Gemini 4 Argon

The First Gemini 4 Model

Google's flagship reasoning model for coding, agents and cyber defense.

1M
token context, input + output shared

1M
output with long decode, up from 64K

#1
Vals Index at 68.9%

Capability map | Benchmarks | API code | Cheat sheet

BUILT FOR

Long agent jobs

Migrated **800K+ lines** of C/C++ to Rust inside Google.

ROLLING OUT

Defenders first

Cyber teams, then paid API and AI Ultra, then everyone.

Swipe

Everything Argon Can Do

One model, ten headline capabilities.

Deep Reasoning
thinking: **low / med / high**

Agentic Coding
DeepSWE **77.9%**, *state of the art*

Knowledge Work
#1 *Vals Index, legal, finance*

IM Context
1,048,576 *tokens in + out*

GEMINI 4
Argon
gemini-4-argon

IM Output
long decode, up from **64K**

Multimodal In
text, image, video, audio, PDF

Agentic Video
LVBench **91.7%**, *long videos*

Tools & Grounding
functions, Search, Maps, code exec

Computer Use
preview · Agent's Last Exam **39.5%**

Cyber Defense
finds & patches vulns · CWE **68%**

Structured outputs · *File search* · *URL context* · *Robotics*

The Spec Sheet

What You Get on Day One

A new frontier model, not a fine-tune of Gemini 3.

MODEL ID
`gemini-4-argon`
plus `-cyber-permissive`

ARCHITECTURE
Sparse MoE
transformer, trained on TPUs

INPUT
Text, image, video, audio, PDF

OUTPUT
Text only
no image or audio generation

CONTEXT WINDOW
1,048,576
shared by input, thinking, output

MAX OUTPUT
65,536 → 1M
default → with long decode

THINKING
LOW / MEDIUM / HIGH
MEDIUM is the default

KNOWLEDGE CUTOFF
March 2026
pair with Search grounding

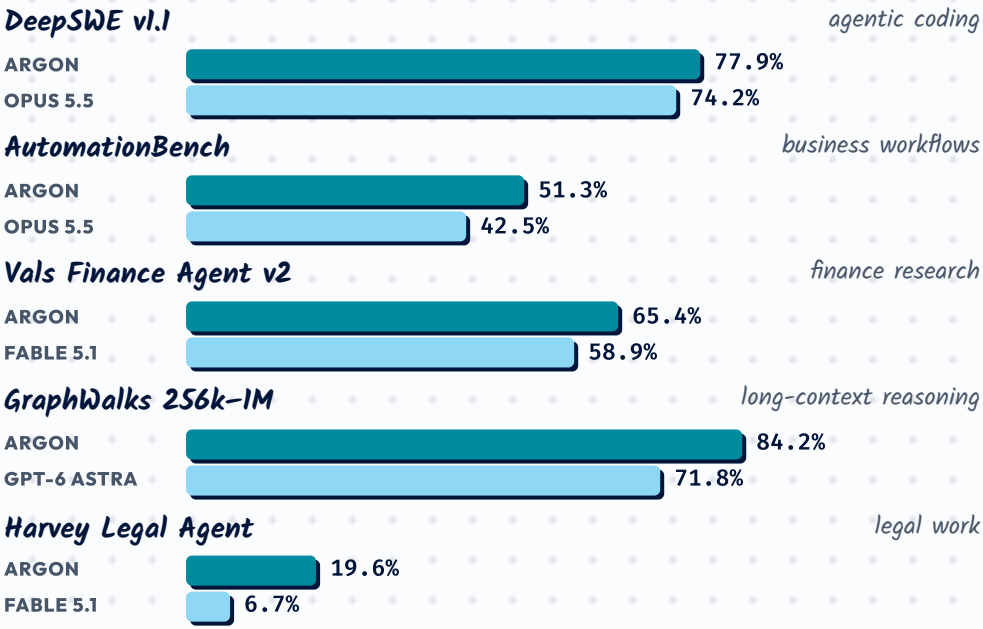
NOT SUPPORTED
Live API · explicit caching · audio and image output

Thinking tokens count against the same 1M budget.

Where Argon Leads

Argon vs the Best Rival

Google's model card, October 2026. Argon in teal.

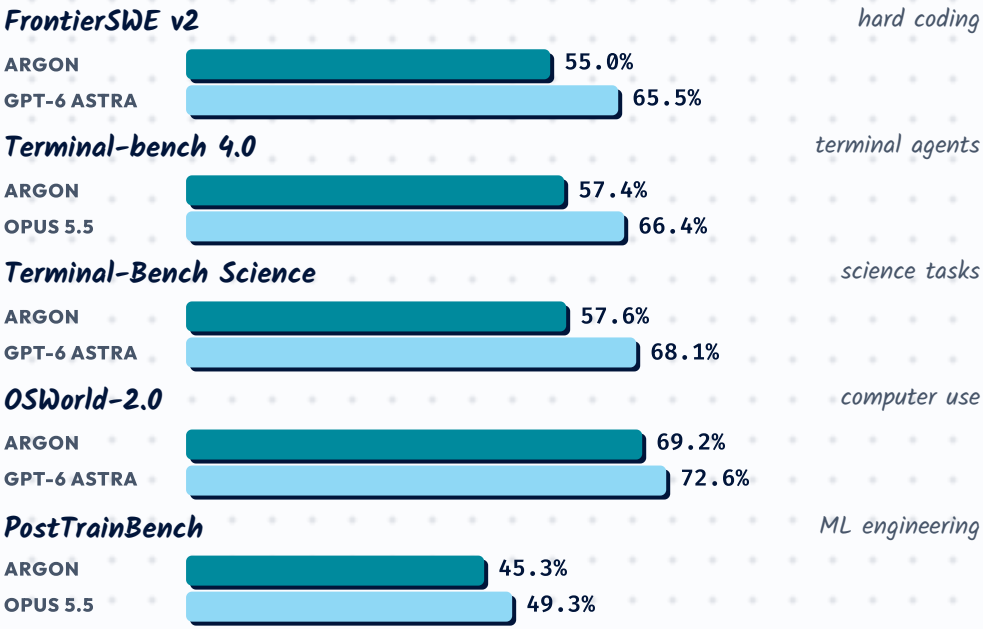


Biggest gaps: **legal**, **automation** and **very long context**.

Where It Doesn't Lead

Strong, but Not a Sweep

Same model card. Other models still win these rows.



*Reported scores use **High** thinking. The default is Medium. Known limits: hallucinations, slow responses, timeouts.*

💡 *Test on **your** tasks before you switch.*

Pick a Thinking Level

One Knob Controls Depth

`thinking_level` *replaces* `thinking_budget`.

HIGH deepest
Multi-file engineering, formal proofs
Most tokens, slowest. Benchmarks were run at this level.

MEDIUM default
General coding and tool use
The default on gemini-4-argon. Start here.

LOW fastest
Classification and extraction
Fewest tokens. Good for high-volume calls.

These now fail or are ignored

- *thinking_budget → HTTP 400*
- *MINIMAL level → HTTP 400*
- *temperature, top_p, top_k ignored*
- *penalties, candidate_count → 400*

Control output with ***thinking_level*** *and* ***response_schema****.*

IM Tokens In, IM Out

The Unified Token Budget

Input + thinking + function calls + answer share one IM budget.

Input
repo, docs, video

+

Thinking
counts too

+

Output
65K default, up to IM

99.7% GraphWalks up to 128k tokens

84.2% GraphWalks at 256k–IM tokens

LONG DECODE · PYTHON

```python
for chunk in client.models.generate_content_stream(
    model="gemini-4-argon",
    contents="Migrate this module to Rust …",
    config=types.GenerateContentConfig(
        max_output_tokens=256000,  # >65,536 = long decode
    ),
):
    print(chunk.text, end="")
```

*The SDK resends with a **continuation_token** for you. Needs **vIbeta**, streaming and google-genai ≥ 2.30.1.*

💡 *Use long decode only when the job needs it.*

Your First Argon Call

Thinking Level + Typed Output

STRUCTURED OUTPUT · PYTHON

```python
from google import genai
from google.genai import types
from pydantic import BaseModel
class Review(BaseModel):
    bottlenecks: list[str]
    fixes: list[str]
client = genai.Client()
resp = client.models.generate_content(
    model="gemini-4-argon",
    contents="Review this order pipeline ... ",
    config=types.GenerateContentConfig(
        response_mime_type="application/json",
        response_schema=Review,
        thinking_config=types.ThinkingConfig(
            thinking_level="HIGH"),
    ),
)
print(resp.text)  # JSON matching Review
```

`schema` *Pydantic class becomes the JSON contract*

💡 *No temperature. Steer with* ***thinking_level*** *and* ***schema****.*

Sees, Hears, Reads

Multimodal In, Text Out

Charts, long videos, audio and whole PDF sets in one prompt.

LONG VIDEO

LVBench 91.7%

State of the art at finding details in long videos.

CHARTS

Chartography 71.6%

Professional chart analysis. Rivals: 46–71%.

DOCUMENTS

PDF sets

Read a series of documents, then act on them.

AUDIO

Speech in

Audio input. No audio or image output.

AGENTIC VIDEO · CONFIG

```
media_processing="agentic"  # default: "static"
```

Agentic mode lets the model inspect a video step by step instead of in one static pass. It is off by default.

*Scenario: drop in a 2-hour incident recording, ask "**when did it break?**"*

Built for Agents

Tools It Can Call

Ground
- Google Search
- Google Maps
- URL context
- File search

Act
- Function calling
- Code execution
- Computer use (preview)

Run
- Batch API
- Flex
- Priority
- Implicit cache

Agent rules from Google's guide

- Match each FunctionResponse to its call `id`, name and order
- Prefer sequential tool turns
- Compact on `initial_prompt_token_count`
- Never end chat history with a `model` turn

Zapier AutomationBench: **#1 at 51.3%.**

Cyber Defense & Safety

Built to Find and Patch Vulnerabilities

It finds, validates and patches critical vulnerabilities on its own.

CWE-bench
68%, tied first with GPT-6 Astra

Fairwind
first rollout to trusted cyber defenders

Permissive
variant without cyber guardrails for vetted teams

Real case: Wiz's Scan for Good used Argon to find a critical bug exposing patient data in healthcare software.

Safety layers

1. **Prompt-injection resistance**
   Google's most resilient model yet
2. **Chain-of-thought monitoring**
   execution stops on misaligned actions
3. **Misuse refusals**
   cyber and CBRN, dual-use research still allowed

*Unjustified refusals fell **9.6 points** vs Gemini 3.1 Pro.*

What Google Did With It

Results From Inside Google

Self-reported numbers from the launch post.

800K+	**C/C++ to Rust migration** *lines ported, up to the Fuchsia Zircon kernel*
2.7x	**Faster video decoder** *libgav1: 32K SIMD lines replaced, identical output*
300+ TiB	**Memory freed fleet-wide** *optimizations found by fleets of agents*
40%	**Better quantum algorithm** *beat a published baseline on qubits × gates*
87%	**Argon for Robotics** *ERQA spatial reasoning. Drove a Spot robot*

A theme: **long, multi-step jobs** that run for hours.

Price, Access, Migration

INTRO PRICE / 1M	LATER / 1M	CACHED INPUT
$2 / $10	$4 / $20	95% off
input / output	input / output	implicit, 6,144+ tokens

Rollout: cyber defenders first, then paid API and AI Ultra, then everyone. Check current access in AI Studio.

From gemini-3.1-pro in 6 steps

1. Change the model to `gemini-4-argon`
2. Upgrade to `google-genai ⩾ 2.30.1`
3. Swap `thinking_budget` for `thinking_level`
4. Drop explicit caching. Put static text first
5. Remove temperature, top_p, top_k, penalties
6. Remove prefilled model turns from history

💡 Static prompt prefix first = cheaper **implicit cache** hits.

Cheat Sheet: Argon Settings

If Your Task Is...

Classify or extract *tags, routing, field pulls*	LOW
Everyday coding, tools *bug fixes, agents with tools*	MEDIUM
Multi-file or proofs *migrations, architecture*	HIGH
Very long answers *full ports, long reports*	long decode
Long video Q&A *recordings, demos*	agentic video
Vuln hunting *vetted defenders only*	cyber-permissive
Typed results *feed downstream code*	response_schema

One knob: thinking_level. One budget: IM tokens.

Sources: blog.google Gemini 4 Argon launch, DeepMind model card, ai.google.dev model page, Google Cloud developer guide, Google Developers Blog (robotics). Benchmarks are vendor-reported, Oct 2026.

AI Coach John FOLLOW | PROITBRIDGE Strive For Better Future
REPOST